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Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
 Seventh Semester B.Tech Degree (S, FE) Examination May 2023 (2019 Scheme)

Course Code: EET401

Course Name: ADVANCED CONTROL SYSTEMS

Max. Marks: 100

Duration: 3 Hours

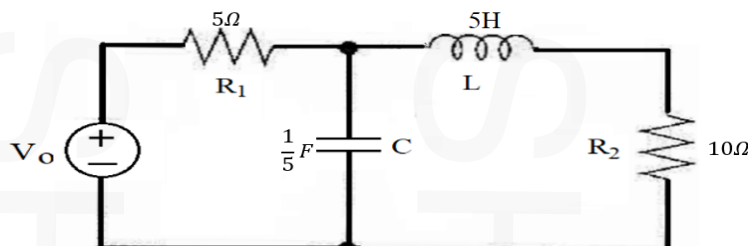
PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|--|-----|
| 1 | Summarize the advantages of state space method over transfer function approach. | (3) |
| 2 | What are phase variables? What are the advantages of choosing phase variables for state space modelling? | (3) |
| 3 | Derive a relation between state model and transfer function of LTI system. | (3) |
| 4 | Write the properties of state transition matrix of discrete time system. | (3) |
| 5 | Define controllability. Explain how can we check the controllability of a system using Kalman's test. | (3) |
| 6 | Illustrate the concept of duality referred to controllability and observability. | (3) |
| 7 | Describe the phenomenon of frequency entrainment in nonlinear systems. | (3) |
| 8 | Derive the describing function of ideal relay nonlinearity. | (3) |
| 9 | Explain what is phase trajectory and phase portrait. | (3) |
| 10 | Determine whether the given quadratic form is positive definite.
$x_1^2 + 3x_2^2 + 11x_3^2 - 2x_1x_2 + 4x_2x_3 + 2x_1x_3$ | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- 11 a) Obtain the state equation for the network shown in Fig. (7)



- b) Derive the state model of field-controlled dc servo motor. (7)

OR

- 12 a) Develop the state space representation of the system with differential equation (7)

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 10y = 8u(t).$$

- b) Determine the diagonal canonical representation of the system with transfer function (7)

$$T(S) = \frac{2(s+5)}{(s+2)(s+3)(s+4)}$$

Module II

- 13 a) A system is described by following state equation. Find the solution of the state (7)

$$\text{equation. } \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

- b) A discrete time system has the transfer function $\frac{Y(z)}{U(z)} = \frac{4z^3 - 12z^2 + 13z - 7}{(z-1)^2(z-2)}$ (7)

Determine the state model of the system in phase variable form.

OR

- 14 a) Find the state transition matrix using Cayley-Hamilton theorem for the system (8)

$$\text{matrix given below } A = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}$$

- b) Write the transfer function of the system whose state model is given by (6)

$$\dot{x} = \begin{bmatrix} -3 & 1 \\ -2 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \quad y = [1 \quad 0] x$$

Module III

- 15 a) Check whether the given system is observable using Gilbert's test. (10)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u$$

$$y = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- b) What you mean by full order and reduced order observer? Explain. (4)

OR

- 16 a) Evaluate the controllability and observability of the state model using PBH test. (8)

$$\dot{x} = \begin{bmatrix} -5 & 1 \\ -8 & 1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [2 \quad 0] x$$

- b) Illustrate the pole placement technique used for control system design. (6)

Module IV

- 17 a) Define Describing function. Explain how describing function can be used for stability analysis of nonlinear systems. (6)
- b) Develop the describing function of relay with dead zone nonlinearity. (8)

OR

- 18 a) Explain any three types of non-linearities that occur in electrical systems. (6)
- b) Derive the describing function of saturation nonlinearity. (8)

Module V

- 19 a) What are singular points? Write about the classification of singular points. (6)
- b) Describe the Lyapunov's Stability criterion and investigate the stability of the following non-linear systems using Lyapunov's method. (8)

(a) $\dot{x}_1 = -3x_1 + x_2$,

(b) $\dot{x}_2 = -x_1 - x_2 - x_2^3$

OR

- 20 a) Determine the stability of the system described by (10)

$$\dot{x} = Ax, \text{ where } A = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}$$

Solve for matrix P in the equation $A^T P + P A = -Q$, assuming the matrix Q to be identity matrix

- b) Define the terms (i) stability (ii) asymptotic stability (iii) asymptotically stable in the large (iv) instability. (4)
